



CS 456: Distributed Databases

Course Code	CS 456
Course Name	Distributed Databases
L-T-P-C	3-0-0-3
Prerequisites	CS204 and CS206
Intended for	Third Year and Fourth Year CS students
Distribution	Elective
Semester	Odd or Even
Approval	4th Senate (May 2013)

Course Contents

(if possible, separated in to approximate units each corresponding to 10 Lectures):

- **Introduction**
 - a. Distributed Data Processing
 - b. Concepts of Distributed Database Systems
- **Distributed DBMS Architecture**
 - Transparency Issues
 - Architectural Models
 - Distributed Database Design
- **Distributed Database Design Issues**
 - a. A Fragmentation b. Allocation
 - c. Integrity Constraints
- **Query Processing**
 - Objectives of Query Processing
 - Layers of Query Processing
 - Query Decomposition and Data Localization
- **Optimization of Distributed Queries**
 - Centralized Query Optimization
 - Join Ordering
 - Distributed Query Optimization Algorithms
- **Advanced Topics**
 - Distributed Transaction Management and Concurrency Control , Distributed DBMS
 - Reliability and Replication Techniques, Multidatabase Systems

Readings

1. M.TamerOzsu, Patrick Valduriez, **Principles of Distributed database systems**, 2nd Edition, Pearson Education.
2. Stefano Ceri, Giuseppe Pelagatti, **Distributed Databases: Principles & Systems**, McGraw- Hill.